

A Dihedrally Equivariant Ordering of Type A Chambers

Let $W = S_n$ be the Weyl group of type A_{n-1} , acting on

$$V = \mathbb{R}^n / \langle (1, \dots, 1) \rangle$$

by permuting coordinates. Let

$$\mathcal{A}_{n-1} = \{ x_i = x_j : 1 \leq i < j \leq n \}$$

be the Coxeter arrangement of type A_{n-1} . Its chambers are naturally indexed by the elements of W , equivalently by the $n!$ linear orders of the coordinates $x_1, \dots, x_n$.

Place the indices $1, \dots, n$ on the vertices of a regular n -gon. This chooses a dihedral subgroup

$$D_n = \langle r, s \rangle \subset S_n,$$

where

$$r : 1 \mapsto 2 \mapsto \dots \mapsto n \mapsto 1$$

is a Coxeter element of type A_{n-1} , and s is a reflection of the n -gon. The group D_n acts on the arrangement, hence on its chambers.

Result. *For every $n \geq 5$, there is a recursively constructed cyclic ordering of all chambers of the type A_{n-1} Coxeter arrangement,*

$$\Phi_n : \text{Ch}(\mathcal{A}_{n-1}) \longrightarrow \mathbb{Z}/n!\mathbb{Z},$$

with the following dihedral equivariance property.

For every chamber C ,

$$\Phi_n(r \cdot C) = \Phi_n(C) + (n-1)! \pmod{n!}.$$

Thus the Coxeter element r acts on the ordering as a rotation of the cycle of $n!$ chambers.

Moreover, the reflection s acts as a reflection of the same cycle: there is a constant $a \in \mathbb{Z}/n!\mathbb{Z}$, depending on the choice of origin of the ordering, such that

$$\Phi_n(s \cdot C) = a - \Phi_n(C) \pmod{n!}.$$

Equivalently, the recursive construction realizes the natural dihedral action

$$D_n \curvearrowright \text{Ch}(\mathcal{A}_{n-1})$$

as the ordinary dihedral symmetry of the cyclically ordered set $\mathbb{Z}/n!\mathbb{Z}$. The ordering is produced recursively from rank $n-2$ to rank $n-1$, rather than by choosing representatives of D_n -orbits.